

# Optimal Athletic Programming in Triathlon using Linear and Integer Programming

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A competitive triathlete must decide:

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**Model 1 (LP)**    $\rightarrow$    **Model 2 (0–1 MIP)**    $\rightarrow$    **Model 3 (LP)**  
training load                      race calendar                      pacing strategy

*Coupled:* Model 1's fitness state gates race readiness and sets the race-day energy budget.

# Real data: 15 months of my own training

- Strava export: **598 activities**, 574 kept (Oct 2023 – Jun 2026)
- Per activity: Banister training impulse  
 $\text{TRIMP} = t \cdot \text{HRr} \cdot 0.64 e^{1.92 \text{HRr}}$
- Exponential smoothing  $\Rightarrow$  fitness CTL ( $\tau=42\text{d}$ ),  
fatigue ATL ( $\tau=7\text{d}$ )
- Current state: CTL = 78.1, ATL = 91.5,  
TSB = -13.4 (fit but fatigued)
- Intensity buckets vs. *sport-specific* threshold HR  
(run 173 / bike 166 / swim 164 bpm)

| $r_{db}/\text{h}$ | easy | mod   | hard   |
|-------------------|------|-------|--------|
| swim              | 43.5 | 94.1* | 136.6* |
| bike              | 68.3 | 86.9  | 142.7* |
| run               | 81.6 | 102.8 | 166.0* |

\* theoretical (sparse data)

# Model 1 — training-load allocation (LP)

**Variables:** hours  $h_{dbw} \geq 0$  per discipline  $\times$  bucket  $\times$  week ( $3 \times 3 \times 16 = 144$ )

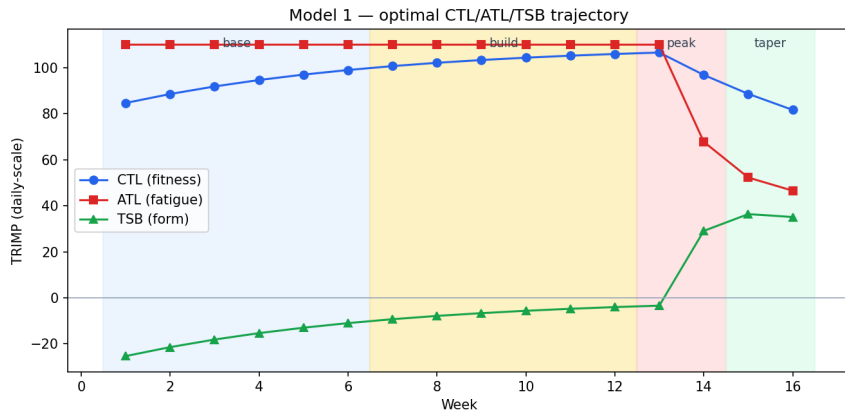
**Banister dynamics as linear equalities** (exact — smoothing is linear in load):

$$L_w = \sum_{d,b} r_{db} h_{dbw}, \quad \text{CTL}_w = \lambda_c \text{CTL}_{w-1} + (1 - \lambda_c) \frac{L_w}{7}, \quad \text{ATL}_w = \lambda_a \text{ATL}_{w-1} + (1 - \lambda_a) \frac{L_w}{7}$$

**Objective:** race-day performance  $\max p_T = k_1 \text{CTL}_T - k_2 \text{ATL}_T$  (TSB-max alone would be degenerate: optimum = rest!)

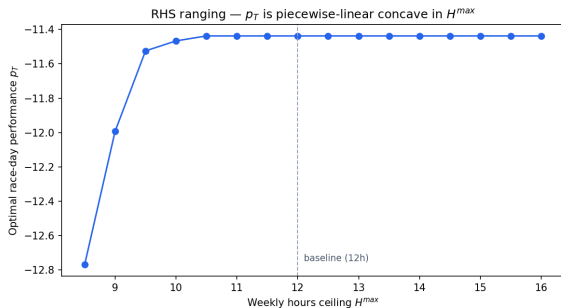
**Constraints:** weekly hour ceilings (phase-dependent)  $\cdot$  discipline min/max  $\cdot$  80/20 polarization  $\cdot$   
 $\text{ATL} \leq 110 \cdot \text{ramp} \leq +10\%/ \text{week} \cdot \text{taper caps}$

# Model 1 — the optimizer discovers periodization



Rides the fatigue ceiling for 13 weeks (CTL 78  $\rightarrow$  107), then tapers. Only weeks 15–16 are *constrained* to taper — **week 14 tapers by choice**. Race day: TSB = +35.

# Model 1 — duality in action



## RHS ranging (hours ceiling 8–16h):

- value function piecewise-linear, concave; slopes  $1.55 \rightarrow 0.94 \rightarrow 0.12 \rightarrow 0$
  - saturates at **10.5 h/week**: the ATL ceiling binds first
- $\Rightarrow$  *recovery capacity, not free time, limits this athlete*

**Largest shadow prices:** race-week minimums ( $-12.9$ /forced run hour in week 16) — the model wants full rest

# Model 1 — alternative optima & heuristic benchmark

**Alternative optima** (degeneracy made visible): fix  $p_T = p^*$ , re-solve min/max total hours

same performance with **124.2h** or **162.3h** of season training — a 38h efficiency gap

**vs. standard coaching heuristic** (fill available hours, 20/45/35 split, 80/15/5 intensity):

- violates the fatigue ceiling in **13 of 16 weeks** (chronic overtraining)
- $p_T = -69.5$  vs. optimal  $-11.4$
- but: heuristic race-day TSB  $+21$  is inside the coaching band  $+15..25$ ; the LP's  $+35$  is not → see  $k_2$  sensitivity

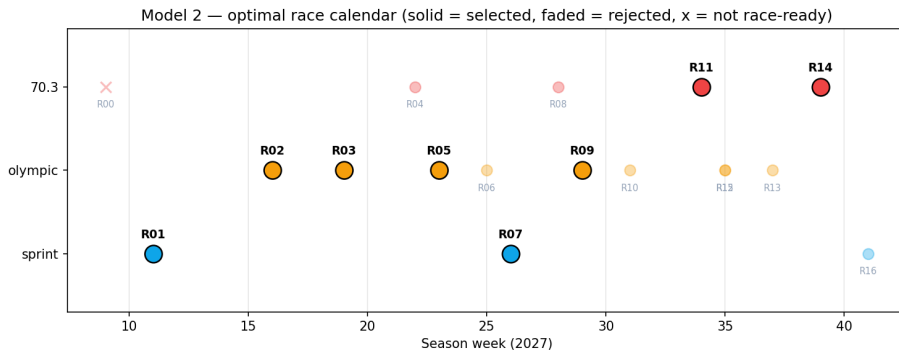
## Model 2 — race-calendar selection (0–1 MIP)

**17 candidate races** (Greek / European / long-haul), binary  $x_i$

$$\max \sum_i q_i \underbrace{(\alpha \text{pts}_i + \beta \text{strat}_i + \gamma \text{enj}_i + \varphi \text{fit}_i)}_{\text{weather-risk-adjusted value}} x_i$$

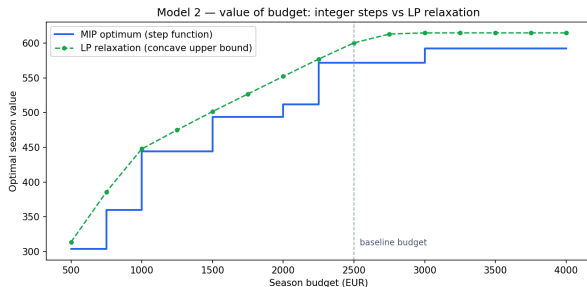
- **Two-dimensional knapsack:**  $\sum \kappa_i x_i \leq 2500$  EUR and  $\sum \delta_i x_i \leq 12$  vacation days
- recovery spacing  $x_i + x_j \leq 1$  (1/2/4 weeks by distance class);  $\leq 2$  races/month;  $\geq 1$  A-race
- **readiness gate from Model 1:** projected CTL must reach the class requirement — excludes a high-value early-season 70.3
- suitability  $\text{fit}_i$  from *objective* course data: elevation gain/km vs. athlete's flat-course preference

## Model 2 — optimal season



8 of 17 races, value 571.7, **2250/2500 EUR and 11/12 days used**. Recovery forces National Champs vs. home 70.3 — optimizer picks the 70.3.

# Model 2 — why Branch & Bound is needed



- LP relaxation:  $z = 600.5$  vs. MIP 571.7 — **integrality gap 5.0%**
- 8 fractional variables; the champs/70.3 conflict splits exactly **0.5/0.5**
- MIP value-of-budget is a **step function** under the concave LP curve
- above 3000 EUR the *vacation-days* budget binds instead — the scarce resource switches
- LP duals: 0.094 value/EUR,  $\approx 0$  per day off

## Model 2 — network-flow extension

Drop the knapsacks and monthly cap  $\Rightarrow$  only recovery spacing remains:

**feasible calendars = paths in a DAG** (nodes: races in week order; arc  $i \rightarrow j$  iff  $\text{gap} \geq \text{recovery}(i)$ )

Three methods, identical result ( $z = 639.4$ ):

- 1 dynamic programming (longest path, topological order)
- 2 **flow LP with continuous variables** — solution integral anyway: network matrix  $\Rightarrow$  *totally unimodular*
- 3 restricted 0–1 MIP

Node–arc incidence matrix  $\Rightarrow$  every basis is a spanning tree: exactly the class *Network Simplex* is designed for (18 nodes, 136 arcs).  $\Rightarrow$  the knapsack constraints are precisely what *breaks* the network structure and forces Branch & Bound in the full model.

## Model 3 — race pacing (LP with linearization)

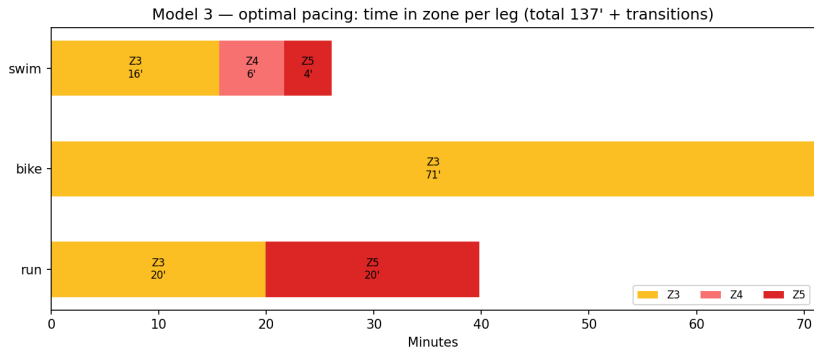
**Variables:** minutes  $t_{\ell z}$  per leg  $\times$  intensity zone (Z1–Z5); 15 variables

**Linearization:** within a zone, speed and energy rate are constants — bike speed  $\propto (\text{power})^{1/3}$  (aero drag), economy degrades with intensity

$$\min \sum_{\ell, z} t_{\ell z} + T_1 + T_2 \quad \text{s.t.} \quad \underbrace{\sum_z s_{\ell z} t_{\ell z} \geq d_\ell}_{\text{distances}}, \underbrace{\sum e_{\ell z} t_{\ell z} \leq k_E \cdot \text{CTL}_T}_{\text{energy budget (Model 1)}}, \underbrace{\text{Z4+ time caps}}_{\text{per leg}}, \underbrace{\varphi \cdot G}_{\text{bike} \rightarrow \text{run}}$$

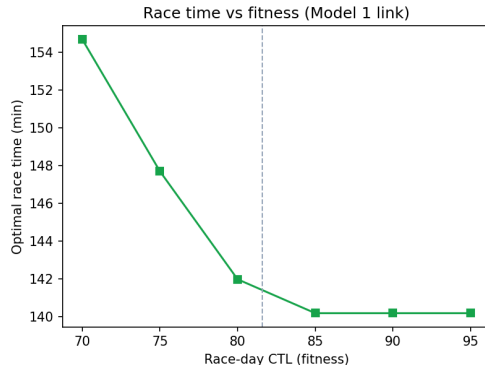
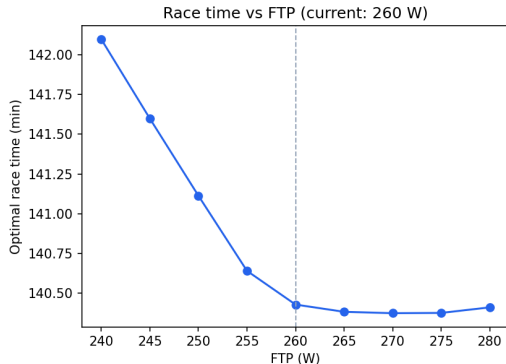
The coupling: each kJ of hard bike work reduces effective run coverage.

# Model 3 — “steady bike, hard run,” derived



Olympic distance in 140.4 min; 9389 kJ budget **exactly** exhausted. Bike entirely at Z3: hard biking buys little speed (cube root) and drains the run.

# Model 3 — what is a watt worth?



Energy dual:  $1 \text{ kJ} = 0.43 \text{ s}$ . Below 260 W each watt  $\approx 6 \text{ s}$ ; **at the current FTP the value collapses** — with a fixed energy budget, more power cannot be fueled. One CTL point = 69–84 s up to CTL  $\approx 85$ .  $\Rightarrow$  *build endurance, not peak power.*

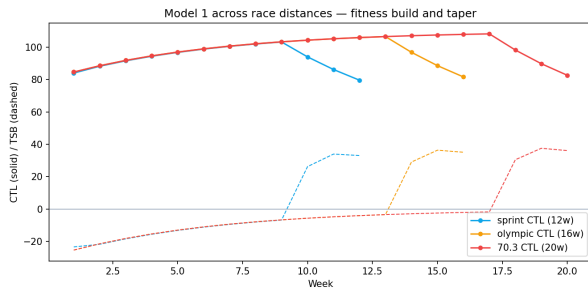
## Model 3 — validated against three real races

| Race                             | CTL  | Swim |      | Bike         |              | Run  |             |
|----------------------------------|------|------|------|--------------|--------------|------|-------------|
|                                  |      | mdl  | act  | mdl          | act          | mdl  | act         |
| Spetsathlon '26 (sprint*)        | 59.0 | 12.9 | 13.2 | 44.4         | 50.9         | 18.9 | <b>17.7</b> |
| Epidavros '25 (Olympic)          | 55.6 | 27.6 | 25.8 | 83.7         | 91.5         | 43.1 | 42.0        |
| 70.3 Costa Navarino '25 (fueled) | 71.4 | 33.3 | 32.9 | <b>181.1</b> | <b>180.3</b> | 86.3 | 115.2       |

Race-day fitness reconstructed from the pipeline;  $k_E = 142$  calibrated on Epidavros. \*non-standard course.

- swim always within 0.4 min; 70.3 bike within **one minute**
- **binding constraints switch with distance**: speed caps (sprint)  $\rightarrow$  energy (Olympic)  $\rightarrow$  *infeasible unfed* (70.3)
- fueling stays LP — fixed rate or *per-leg decision*  $F_\ell \leq \bar{r}_\ell \cdot \text{leg time}$ ; calibrated  $r \approx 21$  kJ/min = **75 g carbs/h, the standard guideline**; absorption duals price **gut training** (+1 kJ/min bike  $\approx 1.7$  min at 70.3)
- the 70.3 run “wall” (4:53  $\rightarrow$  5:52/km, HR 162  $\rightarrow$  145): the energy constraint violated in real life — mechanism right, within-leg timing beyond the model

# Sensitivity across problem sizes and parameters



**Distance scenarios** (12/16/20 weeks, 144–240 vars): longer horizons help ( $p_T$ :  $-13.5 \rightarrow -10.4$ ); build–taper structure invariant

$k_2$  **fatigue weight**  $\in [1, 3]$ : race-day TSB stays  $35 \pm 2$  — taper depth is *constraint*-determined (race-week minimums), robust to the objective's calibration

**Cost ranging** ( $c$ -perturbation): plan invariant for  $k_1 \in [0.68, 1.56]$ ,  $k_2 \in [1.29, 2.95]$  — robust to the least measurable parameters

**Sizes/times:** LP 192 vars / MIP 17 bins / LP 15 vars; all  $<10$  ms (CBC/Simplex)

## Model 2 — the dual problem

LP relaxation is already in standard form  $\max\{c^T x \mid Ax \leq b, x \geq 0\}$ , so the dual is  $\min\{b^T y \mid A^T y \geq c, y \geq 0\}$ :

$$\kappa_i y_B + \delta_i y_D + y_{\text{month}(i)} + \sum_j y_{ij} - [i \in A] y_A \geq v_i$$

*a race enters only if its value covers the shadow cost of what it consumes*

| Dual        | Meaning                                     | Price        |
|-------------|---------------------------------------------|--------------|
| $y_{13,14}$ | Hellenic Champs vs. Costa Navarino conflict | <b>56.40</b> |
| $y_{06,07}$ | Hamburg vs. Thessaloniki conflict           | 31.67        |
| $y_B$       | one extra euro                              | 0.094        |
| $y_D$       | one extra vacation day                      | $\approx 0$  |

⇒ the scarce resource is neither money nor days off but **calendar space**: recovery between races.

# Integrating the pipeline (1): are the objectives coherent?

Model 1 maximizes  $p_T = k_1\text{CTL}_T - k_2\text{ATL}_T$ ; the athlete wants *race time*. Do they agree?

| $\text{CTL}_T$ | $\text{ATL}_T$ | coherent ( $\lambda = 1.5$ ) |              | fitness-only ( $\lambda = 0$ ) |       |
|----------------|----------------|------------------------------|--------------|--------------------------------|-------|
|                |                | $p_T$                        | time         |                                | time  |
| 91.9           | 57.6           | 5.6                          | 163.7        |                                | 140.2 |
| 89.0           | 52.3           | 10.6                         | 142.1        |                                | 140.2 |
| 81.6           | 46.5           | <b>11.8</b>                  | <b>140.2</b> |                                | 140.2 |
| 75.4           | 44.4           | 8.8                          | 147.6        |                                | 140.2 |

- default budget  $E = k_E\text{CTL}$  ignores  $\text{ATL} \Rightarrow$  **all four plans give the same time** — Model 3 is blind to training
- form-based  $E = E_{\text{base}} + k_E(\text{CTL}_T - \lambda\text{ATL}_T)$  is affine in  $p_T$  **iff**  $\lambda = k_2/k_1 \Rightarrow \arg \max p_T = \arg \min$  time
- $E > 0$  needs  $\lambda < \text{CTL}_T/\text{ATL}_T = 1.75$ , so the assumed  $k_2 = 2$  is *incoherent*; **cost ranging** says the plan is invariant on  $[1.29, 2.95]$  — so  $\lambda = 1.5$  fixes it **changing no variable**

## Integrating the pipeline (2): the circular dependency

Model 1 needs the A-race (sets macrocycle length); Model 2 needs Model 1's load (drives the readiness gate). Iterate instead of assuming an order:

| Iter. | Macrocycle | Weeks | Selected A-race           | Outcome            |
|-------|------------|-------|---------------------------|--------------------|
| 1     | olympic    | 16    | R14 Costa Navarino (70.3) | re-plan as 70.3    |
| 2     | 70.3       | 20    | R14 Costa Navarino (70.3) | <b>fixed point</b> |

- converges in **2 iterations**
- exposes a real inconsistency: the calendar picks a **70.3** A-race, which needs a **20-week** build — not the 16-week Olympic macrocycle originally assumed
- the fixed point is Costa Navarino — the race actually contested in 2025

# What the models actually recommend

- **Train more, much easier.** Actual: 6.6 h/wk at **37%** hard. Optimal: 9.0 h/wk at **15%**. Anti-polarised — the one change with the largest payoff.
- **Buying training time is worthless** above 10.5 h/wk (zero shadow price); *recovery capacity* is what binds.
- **Build endurance, not watts.** +1 CTL = 69–84 s; +1 W  $\approx$  6 s below 260 W,  $\approx$  0 above.
- **Ride Z3, run hard, pace evenly.** The 70.3 positive split (4:53  $\rightarrow$  5:52/km, HR 162  $\rightarrow$  145) cost  $\approx$  29 min.
- **Fuel 75 g/h**; gut training worth  $\approx$  1.7 min per kJ/min — at 70.3 only.
- **Protect the calendar, not the budget.** Recovery conflict dual **56.4** vs. 0.094 per euro.
- **Skip early-season Dubai** (not fit in time); if the A-race is a 70.3, build **20 weeks**, not 16.

# Limitations — stated honestly

- **Weekly time step** for Banister dynamics (daily would shift constants, not structure)
- **Sequential pipeline**, not joint optimization — co-optimizing training + calendar is a natural (larger) MIP extension
- **Constructed parameters** in Models 2–3 (race values, zone speeds,  $k_E$ ) — documented assumptions; sensitivity analysis shows which conclusions survive perturbation
- LP's +35 TSB over-taper traced to race-week minimum constraints, not to the model concept

- Three coupled models on **real personal data**, spanning the syllabus: LP + Simplex, duality & sensitivity, IP + Branch & Bound, network flows + total unimodularity
- The optimizer **re-derives coaching wisdom**: periodization, polarized intensity, “steady bike, hard run”
- Duals price the athlete’s true bottlenecks: *recovery capacity* (training), *money* (calendar), *endurance* — *not watts* (race day)
- Fully reproducible: `python -m src.run_all`, 44 tests  
<https://github.com/manos1503/triathlon-optimization>

Thank you! Questions?